

NETRA

Unified CCTV Integration & Intelligence Platform

One federation layer over every camera - reconstruct any vehicle's route across the state, from a single registration number.

Model 1 • Registry + GIS

Model 2 • Analytics

Model 3 • Federation

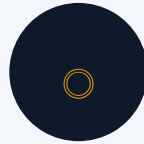


Cameras everywhere - but no way to follow a vehicle



Siloed across departments

Traffic, city police, transport and private VMS platforms don't talk to each other. Footage lives in incompatible islands.



No unified operational view

Operators tab between systems. There is no single map, no shared registry, no cross-system search.



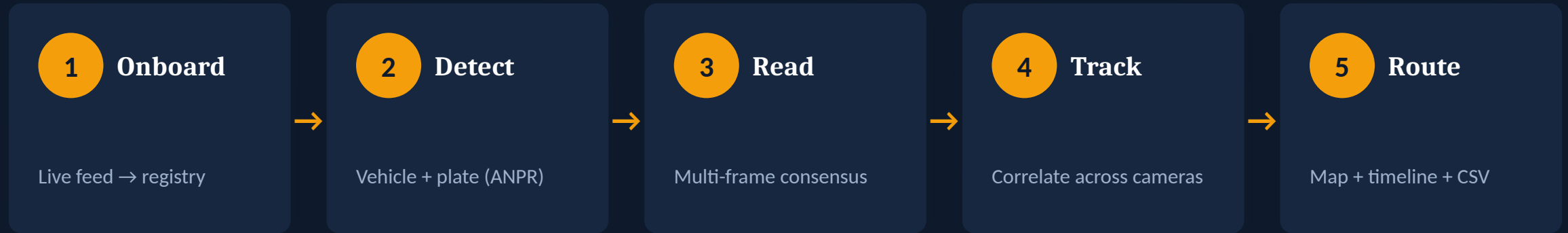
Can't answer “where did it go?”

The core investigative question - a vehicle's path across the city - needs hours of manual review across many recorders.

Netra closes this gap without ripping out a single existing camera or VMS.

THE SCORED TEST CASE

A registration number in - a route out



PROVEN LIVE

GJ01AB1234

vehicle of interest

3

cameras on the route

300

correlated sightings

live RTSP

not a recording

One platform, three models, one federation layer

MODEL 1

Registry + GIS

Every camera catalogued, geolocated, and lifecycle-managed - the single source of truth for the state's CCTV estate.

MODEL 2

Unified viewing + analytics

One map and viewer over all feeds, with ANPR, cross-camera vehicle tracking and evidence-backed alerts.

MODEL 3

VMS federation

Formal adapters normalize independent systems (CSITMS + RTSP/ONVIF) into one event stream with full provenance.

+ selective Model 4 central AI • bound together by a real event-driven federation middleware (Redis Streams → Redpanda at scale)

Adapters → federation → event bus → services



Persistence: PostgreSQL + PostGIS · private object storage (evidence) · signed-JWT auth, RBAC & tamper-evident audit across every layer

The state's CCTV estate, catalogued and alive

33

cameras onboarded

13

districts mapped

100%

geolocated

100%

asset records complete

Live onboarding

Real government feed via a typed adapter; coordinates geocoded and flagged, category inferred - nothing faked.

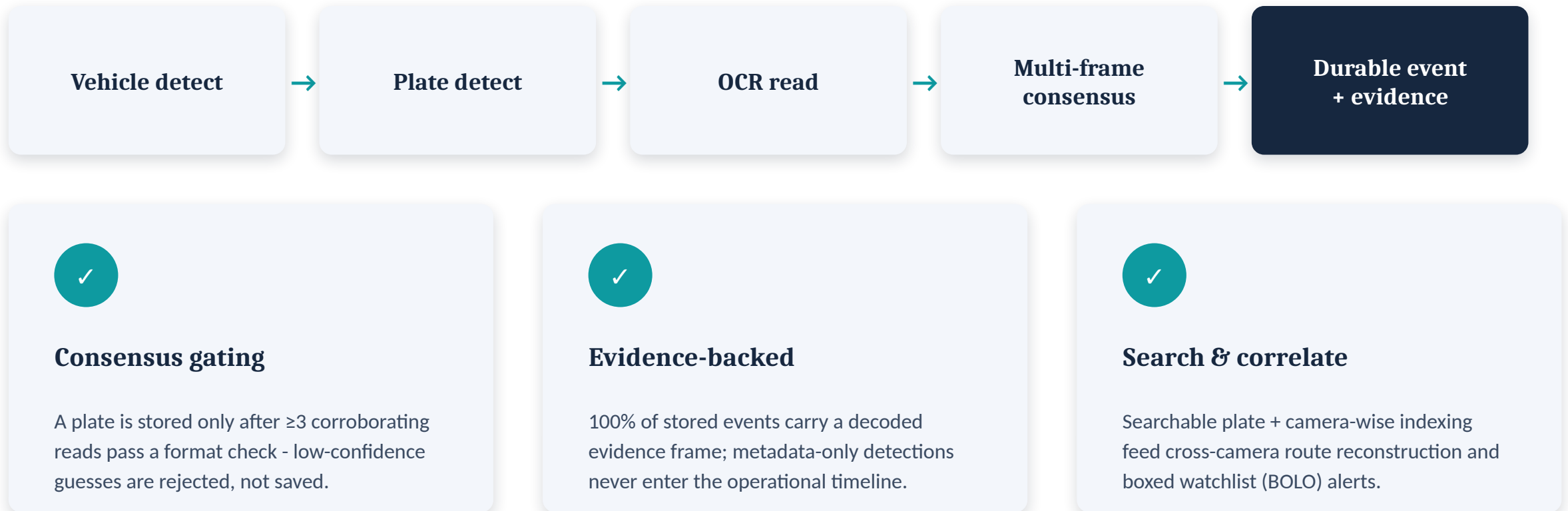
Asset lifecycle

Make, model, install date, AMC and EOL per camera - 17 maintenance-due and 12 end-of-life surfaced automatically.

Coverage & gap analysis

Per-district coverage, thin-coverage and offline lists; downloadable gap report for planning.

Real analytics, evidence-gated end to end



Honest by design: plate yield on wide-angle overview feeds is near-zero (a source-resolution limit), so a controlled ANPR-grade RTSP clip proves the full read → route → CSV path.

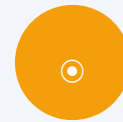
Two genuinely independent systems, one stream



Gujarat CSITMS

Government feed

- Adapter: csitms_api
- Delivery: HLS live edge
- 30 cameras • statewide



Independent RTSP System

Participant / test rig

- Adapter: rtsp_catalog
- Delivery: RTSP (MediaMTX)
- 3 corridor cameras

Same vehicle, both systems. Every event carries source-system provenance; the cross-camera route spans both - federation report shows **demonstration_gap: null**.

Real data, or an honest error - never a fake



Live feeds only

Cameras come only from the real government feed. If it's unreachable, the registry stays empty and the true error is surfaced - no synthetic substitution.



Analytics from the feed

Congestion and surge alerts are computed from actual per-frame counts against a learned baseline - not mock database rows.



Representative data is labelled

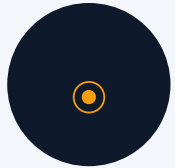
Watchlist, asset lifecycle and demo users are clearly marked representative (challenge Step 3) - never passed off as live government records.



Fail gracefully

Firewalled RTSP, a 502 media plane, an offline feed - each degrades visibly with the real reason, never a silent fallback.

Built for a police-grade chain of custody



Signed-token identity

Spoofable role headers replaced by signed JWTs. A forged role fails verification - a spoofed header now returns 401.



Scoped RBAC

State admin, district officer (district-scoped), viewer. Every camera and stat filtered by verified identity.



Private evidence store

Evidence moved off local disk into a private object-storage bucket, served only through an authenticated proxy.

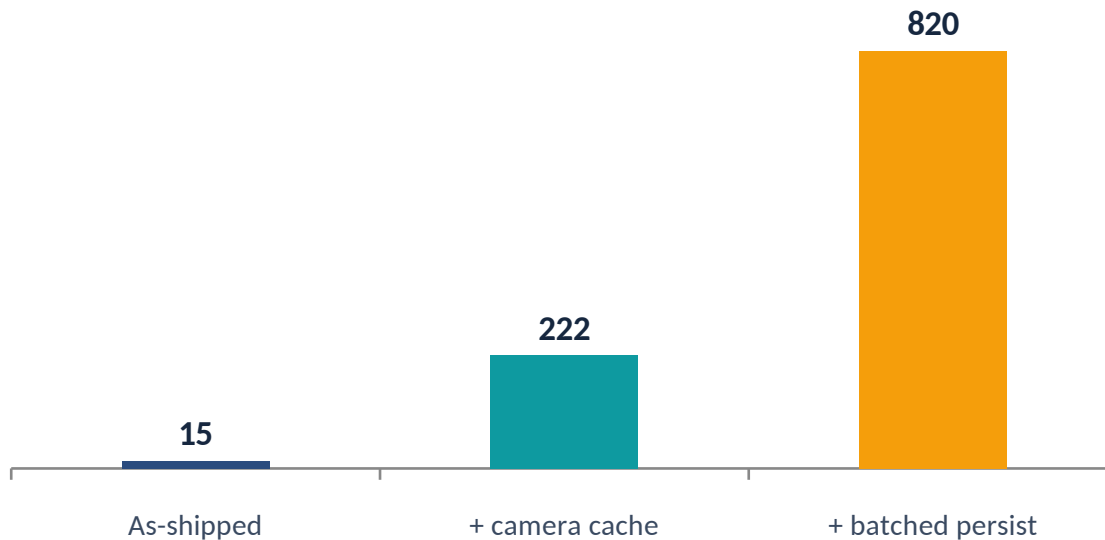


Tamper-evident audit

Every track query, alert ack and BOLO change is attributed to a user and role in an append-only trail.

Benchmarked, bottlenecked, and fixed

Ingestion throughput (POST /api/frame, one backend process)



15 → 820 req/s (~55×)

164 feeds / process

One backend now sustains ~164 feeds at 5 fps - 3.3× the 50-feed target, 0 errors. Replicas scale it linearly.

Analytics is GPU work

Full ANPR is ~5 s/frame on CPU - confirming the design: edge/regional GPU pools (~40 cams/GPU), not a central CPU farm.

Bus is never the bottleneck

Statewide event load (40–200k/s) is ~0.3% of the video data; a small Redpanda cluster clears it with headroom.

All three models - measured PASS

MODEL 1

PASS

Registry + GIS

33 cams · 100% mapped · 100% asset records

MODEL 2

PASS

Unified viewing + analytics

100% events evidence-backed · live ANPR route

MODEL 3

PASS

VMS federation

2 adapters · provenance · demonstration_gap
null

Every non-negotiable blocker closed

✓ ANPR correctness

✓ Analytics stability

✓ Truthful KPIs

✓ Auth: headers → JWT

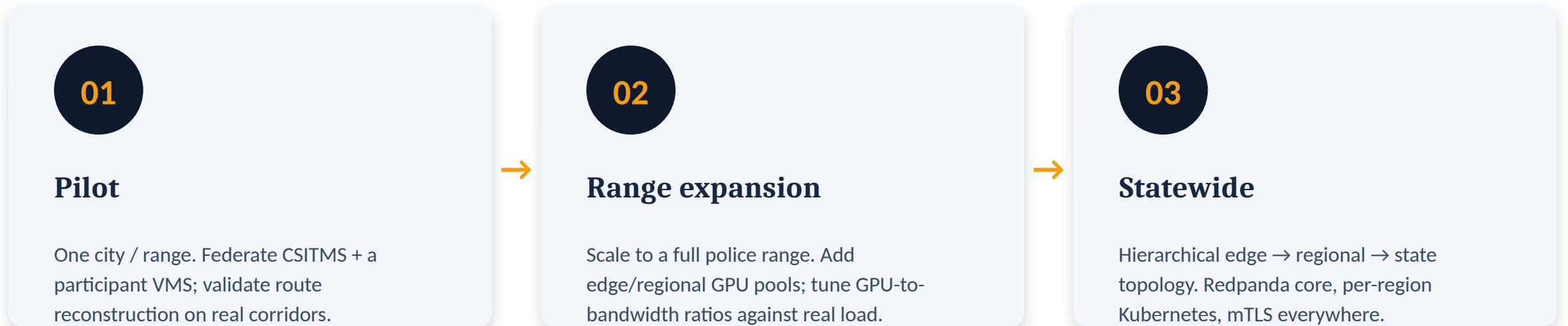
✓ Evidence → storage

✓ 2-system federation

✓ Alert calibration

✓ 50-feed benchmark

From pilot to ~80,000 cameras



Why it's affordable at scale: events are ~0.3% of the data that video is. Cross-camera search runs over ~100 TB of tiny events - not 32 PB of video - so it stays fast and horizontally shardable.



The state's eye - unified.

Model 1 registry + GIS · Model 2 analytics · Model 3 federation - working, measured, and honest.

NETRA